



Problem Statement

Problem

You are given a grid of dimensions $N \times M$. Each cell contains one of the following values:

0 → Empty cell 1 → Fresh orange 2 → Rotten orange

A rotten orange can rot an adjacent fresh orange in 1 unit of time.

A rotten orange at position (i, j) can rot fresh oranges at:

$(i-1, j) \rightarrow (i+1, j) \rightarrow (i, j-1) \rightarrow (i, j+1) \rightarrow$ Right

Find the minimum time required to rot all fresh oranges.

If it is impossible to rot all fresh oranges, print -1.

If there are no fresh oranges initially, the required time is 0.

Submissions

Input Format

The first line contains an integer T , the number of test cases.

For each test case: The first line contains two integers N and

M , representing the number of rows and columns. The next N

lines contain M space-separated integers representing the grid.

Leaderboard

Constraints

$1 \leq T \leq 100$ $1 \leq N, M \leq 500$ $\text{grid}[i][j] \in \{0, 1, 2\}$

Discussions

Output Format

Print a single integer representing the minimum time required to rot all fresh oranges.

Print -1 if some fresh oranges can never become rotten

Sample Input 0

```
3 3
2 1 1
1 1 0
0 1 1
```

Sample Output 0

```
4
```

Sample Input 1

```
3 3
2 1 1
0 1 1
1 0 1
```

Sample Output 1

```
-1
```

Change Theme

Language

Python 3



```
2 from collections import deque
3 n, m = map(int, input().split())
4 grid = [list(map(int, input().split())) for _ in
5         range(n)]
6 fresh = 0
7 for i in range(n):
8     for j in range(m):
9         if grid[i][j] == 2:
10            q.append((i, j))
11        elif grid[i][j] == 1:
12            fresh += 1
13 time = 0
14 while q and fresh:
15     for _ in range(len(q)):
16         i, j = q.popleft()
17         for di, dj in [(-1, 0), (1, 0), (0, -1),
18                       (0, 1)]:
19             ni = i + di
20             nj = j + dj
21             if 0 <= ni < n and 0 <= nj < m:
22                 if grid[ni][nj] == 1:
23                     grid[ni][nj] = 2
24                     fresh -= 1
25                     q.append((ni, nj))
26     time += 1
27 print(time if fresh == 0 else -1)
```

Line: 26 Col: 34



Upload Code as File

Run Code

Submit Code



Test against custom input

✔ Test case 0

✔ Test case 1

Compiler Message

Success